

Firmware versioning guidelines

R&D Squad

V1.0

Introduction

To be able to track firmware migration and spread within our clients, we must ensure reliable and stable devices firmware versioning system. Right now, there are different types of firmware versions which depend on device type. The basic idea is to introduce a firmware build number into firmware version.

This document covers only active and actively developed devices. Devices that are in End-Of-Life state or no longer being developed, should continue with old version scheme.

Of course, devices lists should be updated on a regular basis if any changes occur.

Main devices

Device list:

- ESIM384;
- PT-PRO;
- ET083;
- ESIM320;
- ESIM252.

There are 3 digits used in device firmware versioning. Those should be seen as **(MAJOR).(MINOR).(BUILD_NR)** digits.

Each new planned firmware release should start from new either MAJOR or MINOR digit change, and BUILD_NR should be set to 0.

- MAJOR digit is changed, when there are major functionality changes, totally redesigned major function, a code flow logic change, which is visible to user. New firmware is not backwards compatible with previous ones.
- MINOR digit change should be used in a planned release, functions logic is changed, bugs are fixed, minor changes are added. Firmware remains backwards compatible.

Device configuration change can occur at any (MAJOR or MINOR) version digit change, except the BUILD number digit.

The firmware versioning flow should be as follows:

- In Redmine a new milestone is created, for example v02.03.XX;
- When development is done, in GIT, a release branch with same milestone name should be created, to work on;
- When a firmware is given to QA team for testing for the first time, BUILD number should start at 0;
- Any other succeeding firmware Build, given to QA team for testing, when bug fixing occurs or additional development is done, should have incremented build number;
- Any build number iteration and change should be seen at GIT and must be tracked by release master or masters;
- When firmware is ready for deployment, it is released with the last BUILD number (for example v02.03.15). In GIT, when a push to a master branch occurs, there must be created a tag, with exact FW version;
- Any firmware, which is given for Field-Testing will have its own BUILD number and can be easily tracked;
- If any HOTFIX occurs, a BUILD number must be incremented, to differ from initial release. A push to master branch should be done, a new version tag created.

- When a new firmware version is deployed, support team should be given instructions. In case something goes wrong: what their actions should be and how to solve those issues. For example, if this does not work, roll back to previous firmware version. Also, a support team must be provided with guides, how to use new functions and changed functionality, as their role is to communicate with the user and provide him guidance.
- When a new firmware version is deployed, manufacturing team must be informed. If there is a stock of manufactured devices at warehouse and a firmware update is needed: when new firmware is uploaded to the device it must be reset to default state, to have latest firmware default config.

Peripheral RS-485 devices

All peripheral devices, which do communicate on RS-485 bus:

- ELAN3-ALARM;
- EPGM1;
- EKB2;
- EKB3;

Those devices have only 2 digits for firmware version to store and report to master device. Firmware version 2 digits should be interpreted as **(MAJOR/MINOR).BUILD_NR**. Firmware versioning flow should be the same, as with Main devices, except there is only a single digit for device version. Git flow should be the same.

Peripheral RF devices

Peripheral RF devices:

- EWM1;
- EW2;
- EWF1;
- EWF1CO;
- EWR2;
- EWS2;
- EWS3;
- EWD2;
- EWD3;
- EWK2;
- EWK2A;
- EWP2;
- EWP3;
- EWKB3;
- EWKB4;
- EW1;
- EW2.

All those devices have only 1 digit for firmware version to store and report to main units.

If there is a planned such device firmware development, additional build number implementation into firmware versioning should be thought of. All other and newly developed RF devices should implement 2 digits firmware version indication, as RS-485 peripheral devices do and EWKB5 device.

EWKB5 device

This is RF device and has 2 digits for firmware indication. Firmware version update flow should be the same, as RS-485 peripheral devices.

New devices

All newly developed devices should implement either 3 digits or 2 digits firmware versioning scheme, which depends on device type: peripheral of master/main device.

Future evolution

Future evolution of device firmware version should be in direction of introduction of 4 digits for firmware version or any other version schemes, that will include more usable information. This is to be discussed.

Table of Contents

Introduction	2
Main devices	2
Peripheral RS-485 devices	3
Peripheral RF devices	3
EWKB5 device.....	4
New devices	4
Future evolution.....	4